Ahamed Musthafa R S

Senior Generative AI Engineer

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Senior Generative AI Engineer with 7+ years of software engineering experience and 3+ years building production GenAI and LLM systems. Experienced in Agentic AI, multi-agent orchestration, RAG, LLM fine-tuning, prompt engineering, and enterprise AI across life sciences and customer support. Delivered AI solutions that reduced clinical query-build time by 75%, integration effort by 50%, and platform effort by up to 80%. Strong backend engineering experience with Python, Django/FastAPI, PostgreSQL, and AWS.

WORK EXPERIENCE

Senior Gen AI Engineer

August 2026 - *present*

Tredence Inc, Bengaluru, KA

- Developed a retrieval-augmented multi-agent support workflow using LangGraph and PostgreSQL/pgvector, grounding generated responses in enterprise knowledge while enabling specialized agents for retrieval, reasoning, response validation and escalation.

Senior Research Engineer

June 2021 - July 2026

Saama Technologies, Coimbatore, TN

- Developed an LLM orchestration harness for controlled execution of deterministic AI workflows, incorporating grounded-source validation, compliance checks, and workflow controls to improve reliability of generated outputs.
- Engineered a production-grade Gen AI copilot for Interactive Review Listing (IRL) using LLM APIs and advanced prompt engineering, accelerating clinical query-build time by 75% and improving data extraction efficiency for pharma data-review teams.
- Built a GenAI-powered code generation tool using prompt engineering and Chain-of-Thought reasoning, enabling users to replace complex manual workflows with natural language inputs.
- Fine-tuned open-source LLMs using LoRA on synthetically generated domain datasets, achieving >80% classification accuracy in identifying logical discrepancies within clinical data features.
- Architected ML model service interfaces enabling seamless integration with rule engines, reducing integration time by ~50% and enabling teams to consume model outputs directly.

Backend Developer

July 2019 - June 2021

Zelthy, Bengaluru, KA

- Built full-stack CRM applications using Python, Django, JavaScript, and PostgreSQL for cloud-based therapy management across multiple pharma clients, serving as the engineering foundation for patient support programs.
- Designed and implemented rule logic for Patient Assistance / Support Programs (PAP/PSP) within the Zelthy-Core product, increasing product efficiency by 40% over the prior build.
- Integrated AWS CloudWatch for real-time monitoring of application and network logs, improving observability and reducing mean time to detection for incidents.

EDUCATION

Bachelor of Engineering - Computer Science and Engineering (2015-2019)
Government College of Technology, Coimbatore, Tamil Nadu

HSC - Math/CS (2014-2015)
Reliance Matriculation Higher Sec. School, Namakkal, Tamil Nadu

SKILLS

Gen AI	OpenAI, Cohere, LangChain, LangGraph, Agentic AI, Fine-Tuning (LoRA, QLoRA), RAG, Prompt Engineering, LLM Evaluation
Languages	Python, JavaScript, SQL, NoSQL
Libraries/Frameworks	Pandas, Transformers, Keras, Scikit Learn, Gradio, Streamlit, FastAPI, Django, Flask, SQLAlchemy, Celery, Selenium, BeautifulSoup, Plotly, Boto3
Tools	Windsurf, Claude, AWS (EC2, S3, RDS, Lambda, BedRock, CloudWatch, Boto3), Kubectl, Docker, Airflow, Postman
DB/DW	PostgreSQL, MySQL, MongoDB, Snowflake
Domain	CDISC Standards, Clinical Trials Design, Pharma CRM

PROJECTS

Clinical AI Workbench
https://clinical-ae-workbench.vercel.app/
Developed a lightweight, no-PHI pharmacovigilance AI workbench to simplify public adverse-event exploration, adding PRR/ROR signal metrics, source-grounded summaries, and a faster guided workflow over public FDA data.

DeciSentient - AI Based Decision Agent
https://github.com/amrs-tech/DeciSentient/
Built an autonomous AI decision agent for shopping and travel that combines TinyFish WebAgent for web interaction, Gemini for reasoning and ElevenLabs for voice-based interaction, enabling users to research and compare real-world options conversationally.

CodeProof - Code Analysis Agent
Developed an autonomous code-review and remediation agent capable of analyzing repositories, detecting fragile implementations and technical debt, reasoning about corrective actions, and generating targeted code changes.